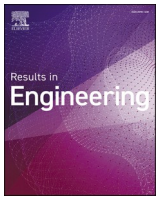
Results in Engineering 27 (2025) 106406

Contents lists available at ScienceDirect

Results in Engineering

journal homepage: www.sciencedirect.com/journal/results-in-engineering



Review article

Closing the artificial intelligence skills gap in construction: competency insights from a systematic review

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ARTICLE INFO

Keywords:
Artificial intelligence
AI skills
Competencies
Construction professionals
Technology integration

ABSTRACT

The integration of Artificial Intelligence (AI) into the construction industry is transforming traditional practices and demanding a new set of competencies from professionals. However, a comprehensive understanding of the specific skills required remains limited. Addressing this gap, a systematic literature review and meta-analysis is conducted in this study to identify and synthesise the AI-related skillsets essential for construction professionals to effectively integrate AI into project delivery. Employing the PRISMA framework and scientometric analysis using VOSviewer, the study analysed 17 key publications sourced from multiple academic databases. The findings reveal eight core AI skill domains: Data Literacy and Management, AI and Machine Learning Fundamentals, Programming and Computation, Digital Collaboration Tools, AI-Driven Decision Making, Cybersecurity Awareness, Change Management and Adaptability, and Communication and Collaboration. The study also highlights a global disparity in research contributions, with notable under-representation from Africa and North America. These insights underscore the urgent need for targeted training, policy development, and collaborative research to bridge the AI skills gap in the construction sector. In addition, the study reveals that, the underdevelopment of AI skills among construction professionals is shaped by a complex interplay of educational, cultural, infrastructural, economic, policy-related, and industry-specific factors. This review provides a foundational roadmap for developing initiatives aimed at equipping construction professionals with the competencies necessary to thrive in an AI-enhanced built environment.

1. Background to the study

Artificial intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, reason, learn, make decisions, and solve problems [1]. Also, it has been defined as the capability of machines to learn, make decisions, and carry out tasks typically associated with human intelligence, drawing from disciplines that integrate concepts from computer science, cybernetics, biology, mathematics, and psychology [2]. AI uses data and algorithms to enable machines to adapt and enhance their performance over time without being explicitly programmed for every task [3,4]. It has been widely used across different industries to improve innovation, accuracy, and efficiency. For example, AI has been applied in the healthcare sector for treatment planning, predictive analytics, and diagnostics [5,6]; for customer service chatbots, algorithmic trading, and fraud detection in the financial sector [7,8]; for process automation, predictive maintenance, and quality control in the manufacturing sector [9]; for customer behaviour analysis, inventory management, and personalised marketing in the retail sector [10]; for traffic management, logistic optimisation, and autonomous vehicles in the transportation sector [11]; for risk management, improving sustainability performance, and project planning in the construction industry [2,12]. These applications demonstrate the versatility and transformative potential of AI across diverse fields.

The construction industry is undergoing a significant transformation driven by rapid technological advancements, with the growing adoption of AI playing a pivotal role in reshaping processes and practices ([13, 14]; CIOB, 2024). In recent years, a significant body of research has emerged exploring the use of AI applications to address various construction industry challenges [12]. These include studies focusing on the impacts of AI on the actualization of project objectives [15], project delay prediction [16,17], performance improvement [14,18,19][], and

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https://doi.org/10.1016/j.rineng.2025.106406
Received 11 June 2025; Received in revised form 12 July 2025; Accepted 20 July 2025
Available online 22 July 2025
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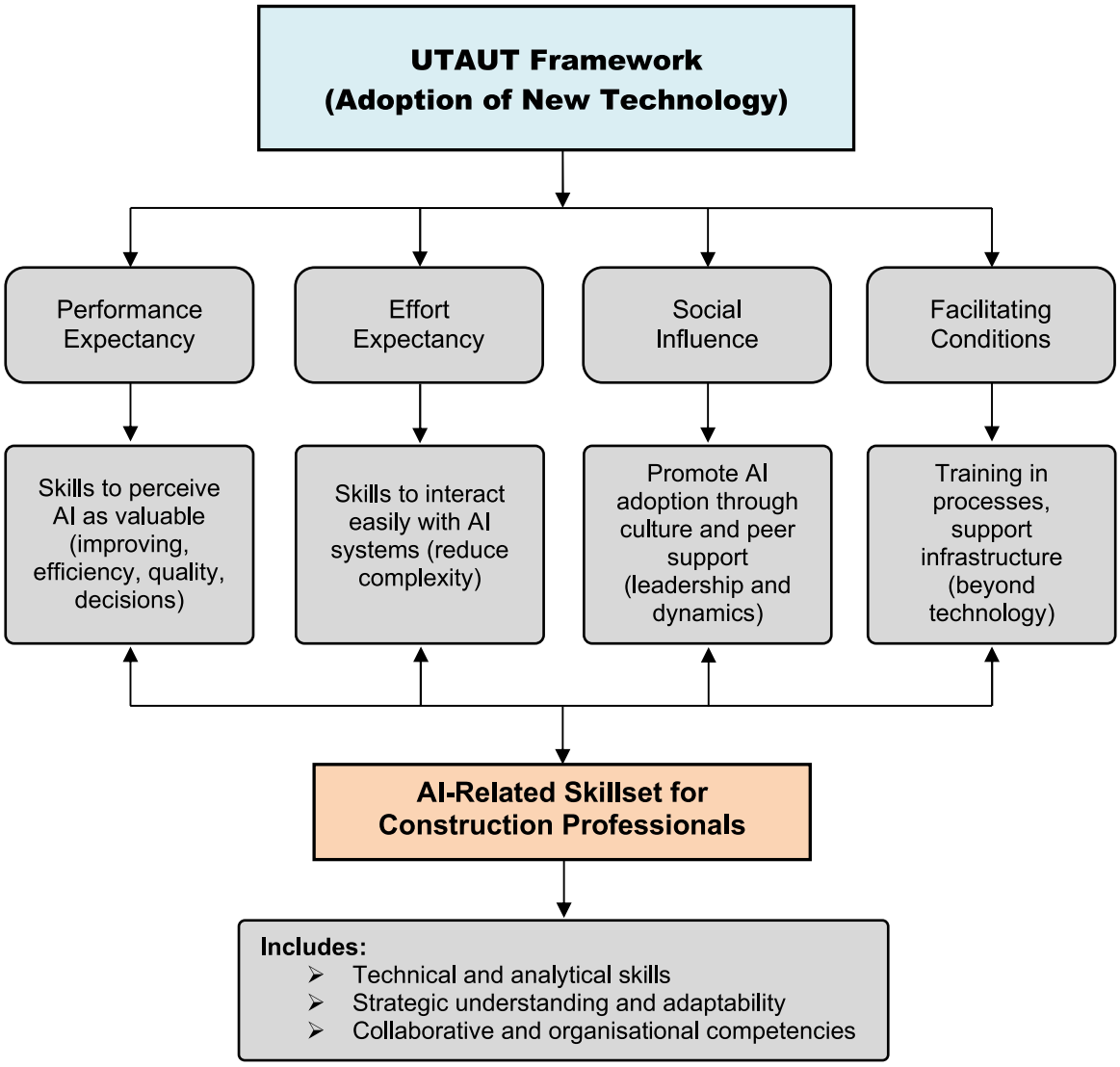

Fig. 1. UTAUT constructs informing AI skills development in construction.

particularly important given the relatively early stage of AI-related skills research in the construction industry. Following a final refinement and abstract screening, 18 documents were selected for detailed analysis. Two publications were removed due to scope of study. Hence, only 18 publications were finalized. The study employed a combination of scientometric review and the PRISMA to examine the selected literature. Fig. 2 illustrates the research design framework. Fig. 2 illustrates the number of records identified, screened, excluded, and included in the final synthesis, along with reasons for exclusions at the full-text stage. The meta-analysis was used to identify the AI-related skills required by construction professionals to effectively adopt and integrate AI technologies in construction project delivery.

The 18 documents retrieved from the Scopus database formed the initial dataset for the scientometric review. This dataset was further refined and 17 documents that met the specific criteria for meta-analysis were selected. Consequently, two interrelated datasets were utilised in this study, as illustrated in Fig. 2.

To broaden the pool of relevant publications for conducting a meta-analysis, additional searches were carried out across other reputable databases, including Web of Science, Google Scholar, and ScienceDirect. The same set of keywords and Boolean operators used in the initial search (“artificial intelligence”, "AI technologies", "construction industry", "skills", "competencies", "professional skills", "technical skills", etc.) were also applied in the additional searches conducted in Web of Science, Google Scholar, and ScienceDirect to ensure consistency in retrieving relevant studies. This yielded 108 additional documents, which, when combined with the 18 papers retrieved from the initial Scopus search, resulted in a total of 126 publications. The application of rigorous inclusion and exclusion criteria – aligned with the predefined research objectives – narrowed this set down to 17 relevant studies. The research workflow, guided by the PRISMA protocol, is illustrated in Fig. 2, while Table 1 and 2 summarizes the selected studies. Each document underwent a comprehensive review to evaluate its relevance to the research objectives, and those that did not adequately address the research objective were excluded. As a result, 109 documents were removed, and the remaining 17 studies were subjected to qualitative content analysis. These selected papers formed the foundation for identifying the AI-related skills required by construction professionals to effectively adopt and integrate AI technologies during construction project delivery.

3.1. Scientometric review

This study employed a technique that enables the visualization and analysis of a large body of literature to illustrate the intellectual progression and structural patterns within a specific research domain.

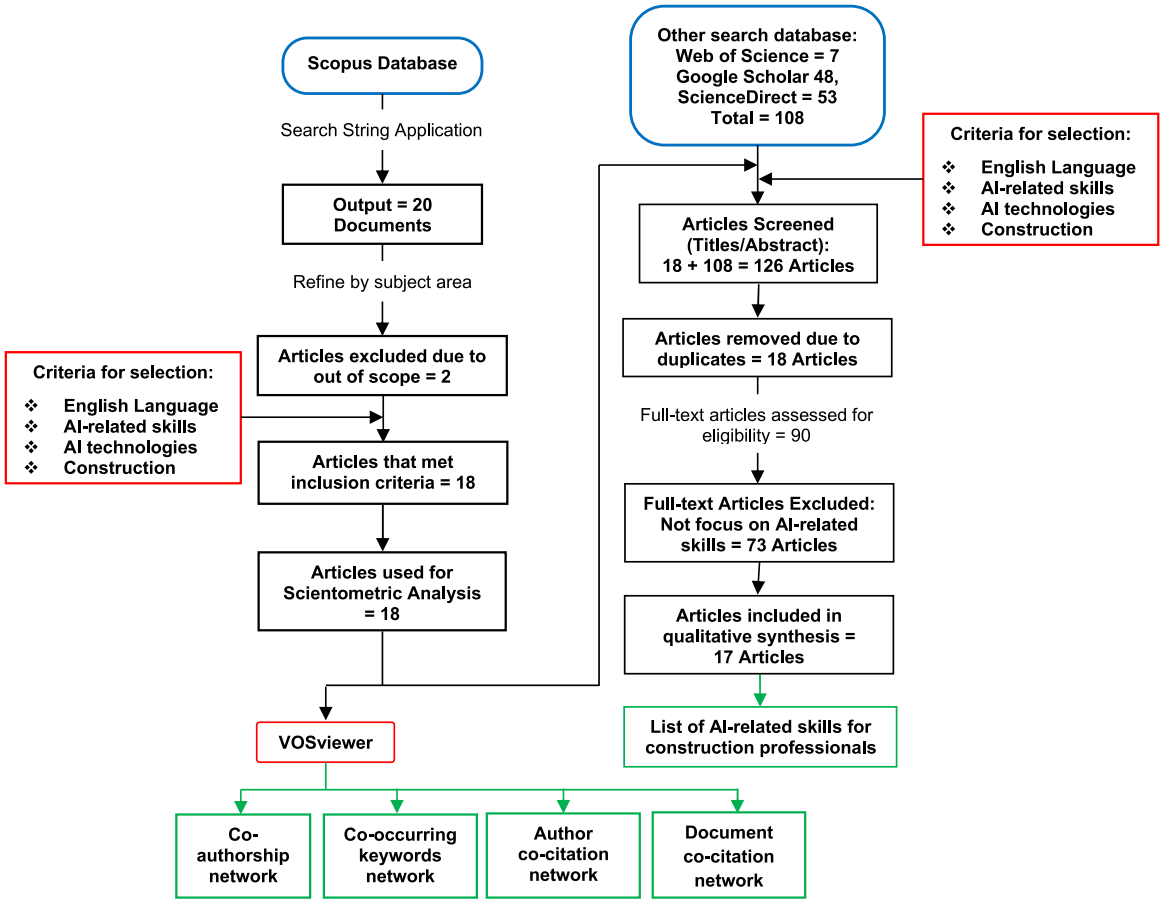

Fig. 2. PRISMA outline of research design.

Although a relatively small number of documents were analysed, reflecting the emerging nature of AI-related skills in construction research. The strength of this approach lies in the critical selection of documents that align closely with the research objectives. This targeted selection offers an advantage over analysing a vast dataset, which may include irrelevant or low-quality papers that could compromise the validity of the findings. Scientometric analysis has gained widespread adoption in various research domains, including early contractor involvement in construction [67], building information modelling [68], green building [69], and sustainability and sustainable development [70]. Other popular review techniques include systematic review, content analysis, and bibliometric analysis among others [57,71,72].

For this study, VOSviewer (version 1.6.20) was used to conduct the scientometric analysis due to its compatibility with Scopus datasets and its established use in related research. The software employs a distance-based visualization approach, where the proximity between nodes indicates the degree of relatedness [72]. By default, it uses association strength normalisation to adjust for variations among nodes. The clustering of networks is performed using the smart local moving algorithm, and items within the same cluster are visually represented by identical colours. VOSviewer also supports both full counting and fractional counting methodologies for network visualization; in this study, the fractional counting method was adopted, as recommended by prior research [73]. The analysis in this study focused on author network, co-author citation, document citation, and co-occurring keywords citation.

4. Results and data analysis

4.1. Annual publication trend and distribution of journals in AI-related research

Though, the studies that applied AI to construction was published in the early 1980s [74]ChenYing, the first paper that discussed AI-related skills in construction dated back to 2017 as shown in Fig. 3. There has been an increase in AI-related skills in construction publications across the globe since the first publication in 2017. The increase in the publications is synonymous with the global growth in AI in construction publications [12,21]; however, the growth in AI-related skills discourse has been rather slow. Focus on AI-related skills in construction research is anticipated to grow steadily, as awareness and adoption of the technologies are still emerging and gradually expanding (Adebowale and Agumba, 2023).

As shown in Fig. 3, 2024 recorded the highest publications in the field of study. Publications are also expected to increase in 2025 as the current study was conducted on April 5th, 2025.

In terms of publication sources, the reviewed studies were distributed across a range of journals, reflecting the interdisciplinary nature of AI skill research in the construction industry (see Table 1). Engineering, Construction and Architectural Management, and Buildings were the most frequent sources, each contributing two studies as shown in Table 1. Other journals, such as Canadian Journal of Civil Engineering, Construction Innovation, Journal of Performance of Constructed Facilities, and Journal of Information Technology in Construction (ITcon), each contributed one study. This distribution suggests that while certain construction-focused journals have begun to emphasise AI skills, the topic is still broadly dispersed across various engineering, technology,

Table 1
Selected research studies for meta-analysis based on AI-related skills needed for construction professionals.

No	Author(s)	Study Location	Focus	Source	Methodology
1	Hanna *et al.* [58]	UK	Automated model that leverages real-world data to determine the weighted competencies of project managers.	Canadian Journal of Civil Engineering	Case study
2	Karki and Hadikusumo [59]	Thailand	Creating an automated model using real-world data to calculate project manager competency weights	Construction Innovation	Qualitative and quantitative approaches
3	Momade *et al.* [44]	Malaysia	AI-based modelling of labour costs	International Journal of Building Pathology and Adaptation	Questionnaire survey
4	Pracucci [60]	Italy	Creating a DT using IoT, BIM, and AI to boost warehouse efficiency and worker safety	Applied Sciences (Switzerland)	Case study, qualitative and quantitative approaches
5	Zhang *et al.* [61]	UK	Drivers of C4.0 tech adoption for improving construction workplace health and safety (CWHS)	Engineering, Construction and Architectural Management	Questionnaire survey
6	Hosseini *et al.* [62]	Iran	Exploring BIM-neural network integration for building element assessment and analysis	Journal of Performance of Constructed Facilities	Case study
7	Peretz-Andersson *et al.* [63]	Sweden	How small and medium-sized manufacturing enterprises coordinate resources for implementing artificial intelligence.	International Journal of Information Management	Case study
8	Knoth *et al.* [43]	Germany	Developing a holistic AI literacy assessment matrix	Computers and Education Open	Conceptual framework
9	Allouzi and Aljaafreh [15]	UK	Prospective impact of AI implementation	Acta Informatica Malaysia	Literature review
10	Regona *et al.* [64]	Australia	Public views AI technologies and explores the potential opportunities and limitations they may present in Australia	Journal of Open Innovation: Technology, Market and Complexity	Qualitative approach
11	Abdullah *et al.* [65]	Malaysia	Construction manager's technical competency for construction Projects	Engineering, Construction and Architectural Management	Interviews and questionnaire survey
12	Balogun [34]	UK	Application of fourth-generation digital technologies in the construction industry	Education + Training	Quantitative approach
13	Souza and Debs [47]	Brazil	Scientifically researched technologies and skills that have been implemented in the construction industry	Buildings	scoping literature review
14	Siddiqui *et al.* [35]	Australia	Creating a clear taxonomy of digital skills essential for implementing Industry 4.0 in construction	Buildings	systematic literature
15	Egwim et al. [16]	UK	Artificial intelligence for predicting construction projects delay	Machine Learning with Applications	Quantitative research
16	Samek *et al.* [66]	Germany	Understanding, Visualizing and Interpreting Deep Learning Models	International Telecommunication Union (ITU) Journal: ICT Discoveries	Literature review
17	Wang et al. [22]	UK	Artificial Intelligence in Cloud Computing technology in the construction industry	Journal of Information Technology in Construction (ITcon)	Systematic review

and education journals. This highlights both the growing interest and the need for more consolidated research efforts in this area.

4.2. Country and regions of publications across the world

Fig. 4 illustrates the geographical distribution of the reviewed papers across various countries and regions across the globe. Europe has the highest number of publications of 10 documents from four major countries (UK, Germany, Italy, and Sweden), followed by Asia with four documents from three major countries (Malaysia, Iran and Thailand); whilst, Australia/Oceania has the third highest publication, with 2 documents from Australia, and the least is South America, with 1 document from Brazil. As shown in the Fig. 4, Africa and North America are the continents that are not represented in the field of study. This indicates a need for enhanced awareness regarding AI-related skills in construction. Construction professionals in those continents need to acquire AI-related skills to effectively integrate emerging technologies, enhance project efficiency, and stay competitive in an increasingly digital and data-driven industry. It is worth noting that while no publications from North America and Africa were identified in this dataset, authors from these regions may still appear in subsequent analyses (e.g. co-citation networks) due to their intellectual influence on the field.

4.3. Visualisation

This section presents the visualisation results of the 18 Scopus-indexed documents, generated using VOSviewer.

4.3.1. Co-authorship network

This network illustrates the collaborative relationships among researchers working in the field of AI-related skills in construction. It highlights how researchers are grouped into clusters based on their areas of interest and co-authorship patterns. A total of 47 researchers contributed to 18 documents. Table 3 lists the leading authors in this domain, while Fig. 5 presents the co-authorship network, which comprises 63 links and total link strength (TLS) of 64. TLS indicates the cumulative strength of an author's connections with other researchers [75].

An overlay visualisation of the network is provided to show both the clustering of authors and the average publication year within each cluster. Notably, the clusters centred on Abbas Bozorgzadeh and Tariq Umar is prominent, reflecting their active research groups and strong collaborative ties with other scholars. The average publication year for these clusters falls within 2023, as depicted in Fig. 5. Additionally, authors such as Altaie and Dishar [76], and Pracucci [60]; represented by red-coloured nodes, are identified as having more recent publications in the field.

The resulting network appears fragmented, loosely connected, and sparsely structured; indicating that AI-related skills in construction research on the continent is still in its early stages. The lack of strong, well-defined connections among many researchers and their respective areas of focus suggest limited collaboration and a developing research landscape.

As shown in Table 3, the paper by Marco Casini [77] recorded the highest number of citations (23), standing out significantly among other authors. This high citation count can be attributed to the paper's

Table 2
AI-related skills for construction professionals identified in previous studies.

No	AI-Related Skills	Author(s)	Frequency
1	Working with AI/IT experts	Allouzi and Aljaafreh [15]	1
2	Data collection, preparation, and visualization	Pracucci [60], Balogun [34], Souza and Debs [47], Siddiqui *et al.* [35], Samek *et al.* [66]	5
3	Data privacy and protection protocols	Souza and Debs [47]	1
4	Using AI insights for scheduling, cost control, and risk mitigation	Zhang *et al.* [61], Allouzi and Aljaafreh [15], Siddiqui *et al.* [35]	3
5	Adapting to AI adoption cycles	Regona *et al.* [64]	1
6	Understanding AI/ML concepts	Karki and Hadikusumo [59], Momade *et al.* [44], Knoth *et al.* [43], Allouzi and Aljaafreh [15], Souza and Debs [47], Siddiqui *et al.* [35]	6
7	Basic Python/R programming	Abdullah *et al.* [65], Souza and Debs [47], Siddiqui *et al.* [35], Samek *et al.* [66]	4
8	Application of predictive analytics	Allouzi and Aljaafreh [15], Egwim et al. [16]	2
9	Managing digital transformation	Zhang *et al.* [61], Peretz-Andersson *et al.* [63], Souza and Debs [47], Siddiqui *et al.* [35]	4
10	Logical thinking and automation setup	Hanna *et al.* [58], Balogun [34]	2
11	Use of cloud-based platforms (e.g., Procore, Revizto)	Souza and Debs [47], Wang et al. [22]	2
12	Understanding AI-related risks	Zhang *et al.* [61], Allouzi and Aljaafreh [15]	2
13	Interpreting AI model outputs	Karki and Hadikusumo [59], Samek *et al.* [66], Egwim et al. [16]	3
14	Translating AI outputs into practical instructions	Abdullah *et al.* [65]	1
15	AI-embedded project management software	Hanna *et al.* [58], Karki and Hadikusumo [59], Hosseini *et al.* [62], Siddiqui *et al.* [35], Samek *et al.* [66]	5
16	BIM data integration with AI tools	Pracucci [60], Hosseini *et al.* [62], Souza and Debs [47], Siddiqui *et al.* [35]	4

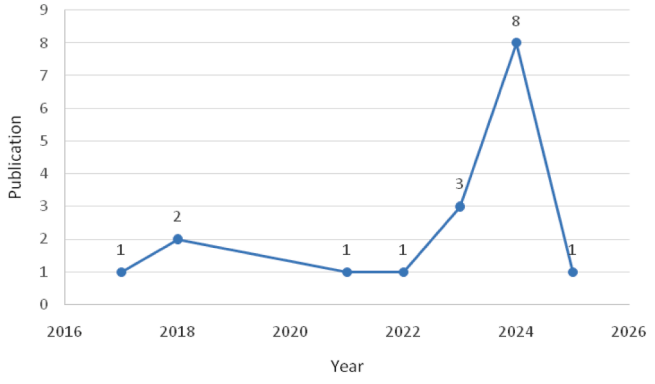

Fig. 3. Annual publication trend of AI-related skills in construction across the globe.

comprehensive coverage of advanced materials, nanotechnology, and smart building systems aimed at improving energy efficiency and environmental performance. Casini's work aligns strongly with current global trends emphasising sustainable and intelligent construction practices, which often involve AI-enabled monitoring and adaptive systems. Consequently, researchers focusing on AI applications in construction, especially in the context of smart and green buildings, have frequently cited this work as a foundational reference. This highlights the interdisciplinary nature of AI-related skills in construction, bridging material innovation, building performance, and intelligent design strategies.

4.3.2. Co-occurring keywords network

This section presents the network of co-occurring keywords extracted from the 18 analysed documents, offering insights into the thematic focus of the research publications [72]. A thesaurus file was used in VOSviewer to merge equivalent keywords (e.g., 'AI' and 'Artificial Intelligence'; 'Optimization' and 'Optimisations'; 'BIM' and 'Building Information Modelling') to avoid duplication and improve clarity of the keyword analysis ([78]BechhoferGoble). The final keywords reflect the aggregated occurrences and total link strength values. A total of 205 keywords were identified across the documents; however, to ensure clarity and relevance, a minimum occurrence threshold of two was applied. As a result, only 26 keywords met this criterion and were visualized in the network shown in Fig. 6. The network comprises 148 links with total link strength of 230. Table 4 highlights the most frequently occurring keywords within the dataset.

The overlay visualization in Fig. 6 illustrates the temporal evolution of research themes based on keyword colour coding. Keywords in blue represent themes with an average publication year around 2018, green keywords correspond to 2022, and yellow keywords signify more recent themes with an average publication year of 2024.

The discourse around keywords such as artificial intelligence, construction industry, construction projects, project managers, managers, and models; are associated with earlier publications, averaging around 2018 and 2020. In contrast, keywords such as forecasting, machine learning, computer vision, innovation, decision making, optimisation, digital transformation, supply chains, building information modelling, and construction professionals reflect a focus on the integration of advanced digital technologies and data-driven approaches in construction processes, with related publications averaging between 2022 and 2024. However, keywords that are centred on AI-related skills or competences for managing construction projects are under explored. This suggests a potential avenue for future research in the field.

4.3.3. Author co-citation network

A total of 2,290 authors were cited across the analysed documents. To ensure relevance, a minimum citation threshold of five was applied, resulting in 27 authors meeting this criterion. Table 5 presents the most frequently cited authors, while Fig. 7 displays the density visualisation of the co-citation network, which comprises 243 links and total link strength of 2135. Together, the table and network visualisation highlight the most influential contributors to AI-related skills in construction research within the context of the study. These frequently cited authors are recognised for their significant impact on the field. The wide geographical spread of these citations underscores the global nature of AI-related skills in construction research [21]. The density map particularly emphasises the prominence of scholars such as David John Edwards from Birmingham City University, UK, and Qian Zhang from Beijing Normal University, China, as key influencers in the domain.

4.3.4. Document co-citation network

This section presents the network of cited references within the analysed documents. It reveals consistent patterns in research themes and highlights frequently referenced works. Table 6 lists the most highly cited documents, while Fig. 8 illustrates the density visualisation of the citation network, comprising 283 links and a total link strength of 829. Across the 18 documents analysed, a total of 1,274 references were cited. To visualise the network effectively, a minimum citation threshold of two was applied, resulting in 35 references meeting this criterion. The most cited documents as shown in Table 6 include studies by Casini [77] and Gerbert *et al.* [79].

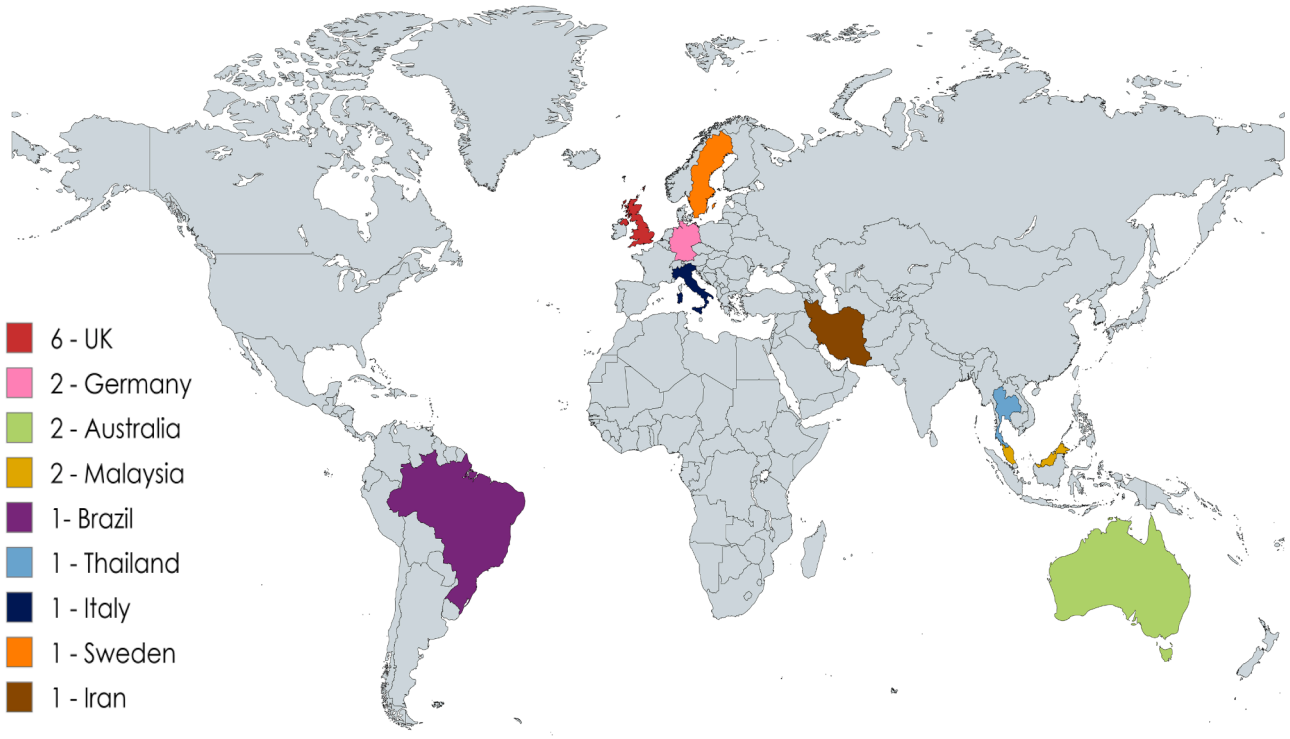

Fig. 4. Country distribution of AI-related skills in construction publications across the globe.

Table 3
Top authors in AI-related Skills in construction publications.

No	Author	Document	Citation
1	Bozorgzadeh, Abbas	2	2
2	Umar, Tariq	2	2
3	Adjei, Elvis Konadu	1	0
4	Adjei-Kumi, Theophilus	1	0
5	Alkali, Abubakar Kori	1	2
6	Altaie, Meervat R.	1	1
7	Beheshti, M.R.	1	0
8	Casini, Marco	1	23
9	Chaabane, Amin	1	0
10	Danso, Anthony Kwame	1	0

4.4. Meta analysis of AI-related skills for construction professionals

The selected studies originate from nine different countries across four continents. The AI-related skills for construction professionals involved in project delivery, as identified in each of the studies, were assigned to sixteen specific AI-related skills (as presented in Table 2). These skills were then thematically grouped into eight broader skill domains and categorised by associated role players within the architecture, engineering, and construction (AEC) industry, as shown in Table 7. The eight skill domains presented in Table 7 were identified through the thematic grouping of the sixteen AI-related skills listed in Table 2. This grouping was carried out through content analysis, where similar or related skills were clustered into broader domains to create a more practical and comprehensive framework for AI skill development in construction. This approach allows for a more structured understanding of skill areas and facilitates targeted training and development strategies.

4.4.1. AI and ML fundamentals skill domain

From the synthesised studies, the AI-related skills that are related to AI and ML fundamentals are the most reported (11 times) in the studies. As an AI skill domain, AI and ML fundamentals cover a basic understanding of how AI and ML technologies work, as well as their core concepts, limitations, and capabilities [4]. For construction professionals, this entails being able to comprehend how AI models are trained, how predictions are generated, and how algorithms can be applied to tasks like cost estimation, schedule optimisation, and risk assessment. In as much as deep technical expertise might not be required, a foundational knowledge enables construction professionals to make informed decisions about adopting AI tools, interpret AI-generated outputs and collaborate effectively with data scientists and technology specialists [83]. This skill domain is vital for leveraging AI to enhance productivity and innovation in construction project delivery.

4.4.2. Data literacy and management skill domain

This is a foundational AI skill domain that allows construction professionals to effectively connect with AI-driven tools and processes. Data literacy and management consists of the ability to collect, clean, organise, interpret, and visualise data relevant to construction projects [84]. As AI systems depend largely on high-quality data for training and decision-making ([85]BalbaaAbdurashidova), construction professionals must understand data sources, formats, and workflows, specifically in terms of tools such as BIM and IoT sensors. Proficiency in data literacy ensures that construction teams can generate actionable insights, improve forecasting accuracy, and support informed decision-making throughout the project lifecycle [86]. This skill domain is necessary for maximising the benefits of AI integration in project delivery.

4.4.3. Digital collaboration tools skill domain

Digital collaboration tools as an AI skill domain entail the effective use of AI-enabled platforms and software that facilitate real-time communication, coordination, and data sharing among construction project stakeholders. Tools like AI-integrated BIM platforms, digital dashboards, and cloud-based project management systems, support faultless collaboration across project teams, even in distributed or remote settings [87]. Construction professionals must be skilled in navigating these tools to manage documentation, resolve issues, track

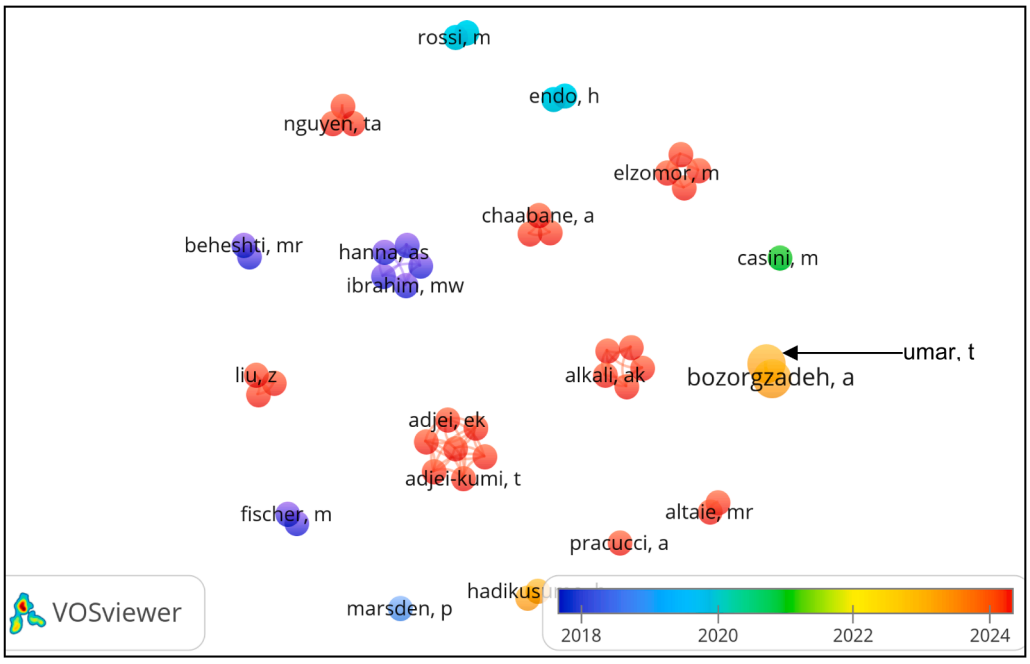

Fig. 5. Co-authorship network of AI-related Skills in construction publications.

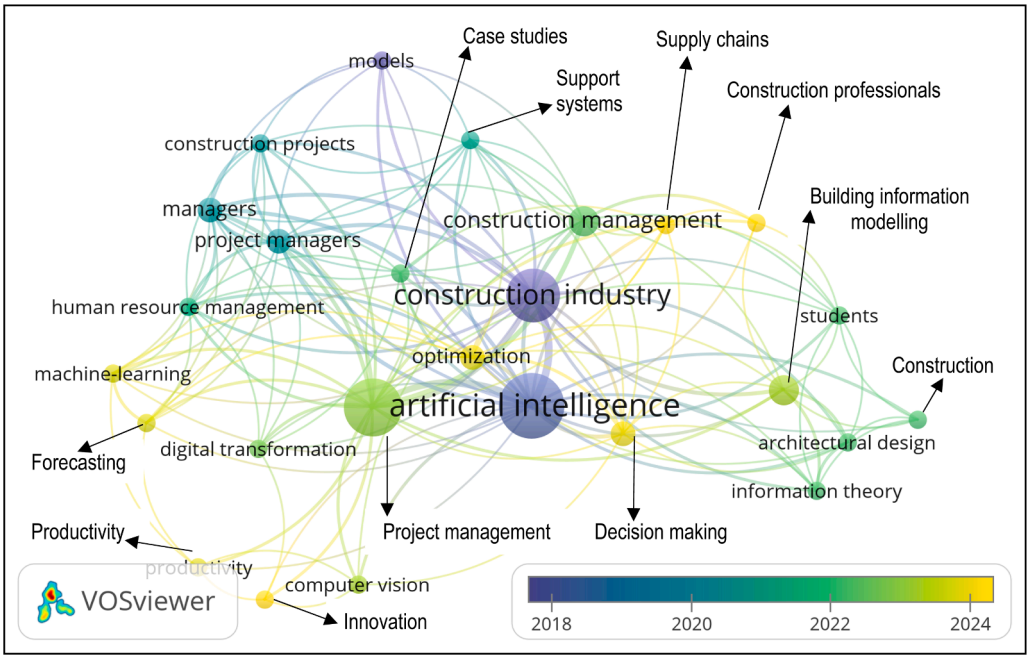

Fig. 6. Co-occurring keywords network of AI-related skills in construction publications.

progress, and assign tasks efficiently. According to Waqar *et al.* [88], mastery of digital collaboration tools enhances transparency, decision-making, and productivity, making it a vital component of AI adoption in modern construction project delivery.

4.4.4. Programming and computation skill domain

This AI skill domain refers to the ability to understand and apply basic coding principles and computational thinking to support the use and customisation of AI tools in construction project delivery. This involves familiarity with programming languages like Python or R, which are commonly used in AI and data analytics, as well as the logic behind algorithm development and automation. This skill enables construction professionals to automate repetitive tasks, interact with AI software at a deeper level, and adapt digital solutions to specific project needs. Even as not every professional need to be a programmer, having a basic grasp of computational concept enhances their capacity to innovate and collaborate in technology-driven project environments.

4.4.5. Change management and adaptability skill domain

This skill domain refers to the ability of construction professionals to effectively navigate and support the transition toward AI-enabled processes and technologies within project environments. This involves understanding the dynamics of organisational change, embracing new digital workflows, and supporting a culture of innovation and

Table 4
Most active keywords of AI-related skills in construction publications.

No	Keyword	Occurrences	TLS
1	Artificial intelligence	12	59
2	Project management	10	47
3	Construction industry	9	47
4	Construction management	4	20
5	Building Information Modelling	4	17
6	Decision making	3	16
7	Managers	3	24
8	Optimization	3	21
9	Project managers	3	24
10	Architectural design	2	12
11	Case-studies	2	13
12	Computer vision	2	8
13	Construction	2	7
14	Construction professionals	2	8

Table 5
Top cited authors of AI-related skills in construction publications.

Author	Institution	Citation	TLS
Edwards, David John	Birmingham City University, UK	14	172
Zhang, Qian	Beijing Normal University, Beijing, China	11	295
Brilakis, Ioannis	University of Cambridge, UK	8	196
Li, Hongyan	Montana State University, US	8	220
Lim, Benson T. H.	University of New South Wales, Australia	8	232
Thwala, Wellington Didibhuku	Walter Sisulu University, South Africa	8	161
Ding, Li	Washington University in St. Louis, US	7	185
Pena-Mora, Feniosky	Columbia University, US	7	198
Zhang, Lei	The Hong Kong Polytechnic University, Hong Kong	7	136

Note: Although no publications from China and the US were identified in the country-level publication analysis, some authors from these countries appeared prominently in the author co-citation network (as shown in Table 5). This is because co-citation analysis is based on the frequency with which authors are cited together in the reference lists of the analysed articles, regardless of whether they themselves authored publications within the dataset. Thus, even in the absence of direct publications from these countries in this specific field, influential authors from China and the US can still have a significant intellectual impact and shape the knowledge structure of the topic under study.

continuous learning. Construction professionals must be open to adopting new tools, adjusting to evolving roles, and guiding teams through periods of transformation. Strong adaptability and change management skills are necessary for ensuring the successful integration of AI in construction, minimising resistance, and maximising the long-term benefits of digital transformation [12].

4.4.6. AI-driven decisionmaking skill domain

This refers to the ability of construction professionals to interpret and apply insights generated by AI systems to foster strategic and operational decisions in project delivery. This includes using AI tools for predictive analytics, schedule optimisation, resource planning, cost estimation, and risk assessment. Construction professionals must understand how to evaluate AI recommendations, assess their reliability, and integrate them into workflows to enhance outcomes. By leveraging AI-generated data and forecasts, construction teams can make faster, more informed decisions, reduce uncertainty, and enhance overall project efficiency and performance [29,89].

4.4.7. Cybersecurity awareness skill domain

This skill domain involves understanding the risks, protection measures, and ethical considerations associated with applying AI and digital technologies in construction project delivery. As AI systems often depend on large volumes of sensitive project and stakeholder data, construction professionals must be aware of potential threats such as data breaches, cyber-attacks and unauthorised access (Nabeel, 2024). This skill includes knowledge of data privacy, protection protocols, and compliance with industry standards. Cybersecurity awareness ensures that construction professionals can safeguard digital assets, maintain trust, and support the secure and responsible implementation of AI tools in construction processes [12].

4.4.8. Communication and collaboration skill domain

This AI skill domain involves the ability of construction professionals to effectively exchange ideas, insights, and information when working with AI tools and multidisciplinary teams. As AI adoption in construction often requires input from project stakeholders, software developers, and data scientists, construction professionals must be able to interpret technical outputs and translate them into actionable plans [12]. This skill includes articulating AI-generated recommendations, facilitating team discussions around data-driven decisions, and ensuring alignment across departments. According to Przegalinska *et al.* [90], strong communication and collaboration enhance the integration of AI into project workflows; promote shared understanding, and support more coordinated and efficient project delivery.

4.5. Factors leading to the underdevelopment of AI skills in the construction industry

The underdevelopment of AI skills among construction professionals remains a critical barrier to widespread technology adoption and digital transformation in the industry. Based on the findings presented in this study (Section 4.2), several key factors contribute to this skillset gap, which vary across different regions. The literature review reveals that gaps in education and training, limited awareness and resistance to change, insufficient technological infrastructure, economic and funding constraints, weak policy and regulatory frameworks, and the construction industry's fragmentation and conservatism are key barriers to AI skills development within the sector [91–104]. The regional disparities highlighted in this study, with Europe leading in contributions and Africa and North America showing none, underscore the urgent need for targeted interventions. Addressing these underlying factors is essential for equipping construction professionals globally with the necessary skills to leverage AI technologies effectively, thereby enhancing productivity, innovation, and competitiveness across the sector. These factors are discussed in the following subsections.

4.5.1. Educational and training gaps

One of the primary reasons for the AI skills gap is the inadequate integration of AI-focused content into both formal construction education and professional development programmes [92,101,102]. In regions such as Africa and parts of Asia, there is a significant lack of specialized courses, workshops, and continuous professional development programs focusing on digital technologies and AI applications in construction. For example, Ukala and Iheukwumere [102] found, among other insights, that the availability of AI tools in Nigeria's Technical and Vocational Education and Training programs was insufficient to effectively support students' learning and skill development. By contrast, countries in Europe, which recorded the highest number of contributions in this field of study (UK, Germany, Italy, and Sweden), tend to have more robust educational frameworks that support research and practical training in emerging technologies [105,106].

4.5.2. Limited awareness and resistance to change

The adoption of technology is significantly influenced by cultural attitudes, which in turn shape skills development [93]. In regions where traditional construction practices dominate, there is often a strong resistance to change and a preference for conventional methods. This is evident in continents such as Africa and South America, where minimal

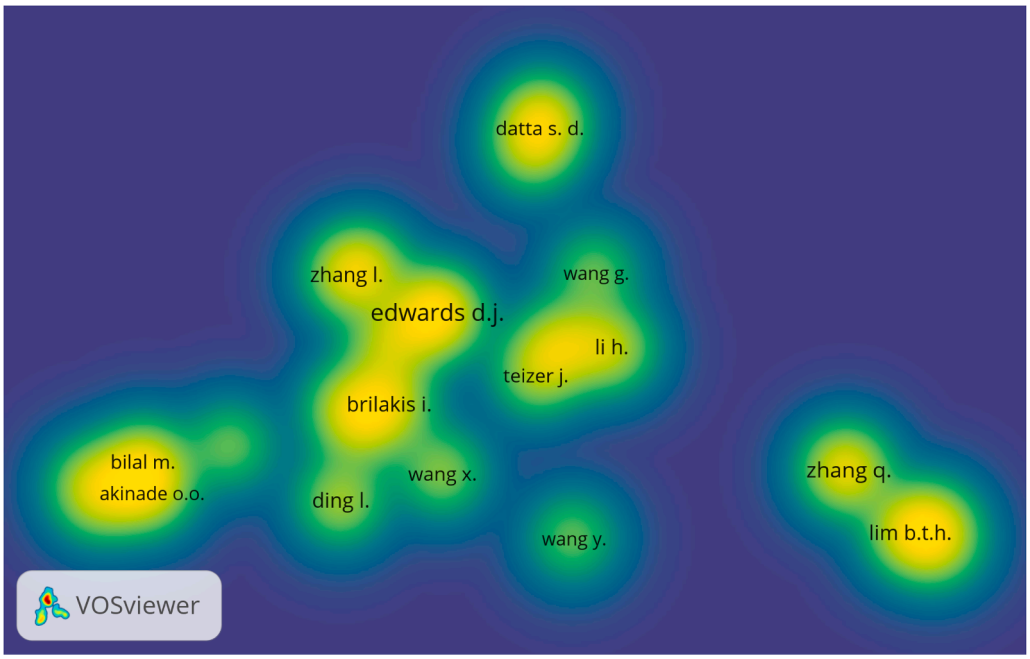

Fig. 7. Author co-citation network of AI-related skills in construction publications.

Table 6
Cited documents of AI-related skills in construction publications.

Document	Title	TLS
Casini [77]	Smart buildings: advanced materials and nanotechnology to improve energy-efficiency and environmental performance	105
Gerbert *et al.* [79]	Digital in engineering and construction, the transformative power of building information modeling	105
Borrmann *et al.* [80]	Building information modeling technology foundations and industry practice	72
Leskinen *et al.* [81]	A review of the impact of green building certification on the cash flows and values of commercial properties	72
Qi *et al.* [82]	A systematic review of emerging technologies in industrialized construction	72

research contributions suggest a lack of engagement with AI tools and methods. In contrast, Europe's higher publication rate may reflect greater openness and proactive efforts to explore and integrate AI solutions.

4.5.3. *Insufficient technological infrastructure*

The availability and accessibility of technological infrastructure greatly influence the development of AI competencies [91,96]. Many developing countries, particularly in Africa, faces challenges such as poor internet connectivity, inadequate digital platforms, and limited access to advanced computing resources [91,107,108]. These constraints hinder both research and practical implementation of AI in construction. On the other hand, countries with more established technological ecosystems, such as those in Europe and Australia [109,110], are better positioned to cultivate and sustain AI skills among construction professionals.

4.5.4. *Economic and funding constraints*

The development of AI expertise requires substantial investment in research, training, and technology acquisition [94,103]. Countries experiencing economic limitations might restrict funding opportunities for both academic institutions and construction firms. This financial barrier discourages experimentation with innovative technologies and reduces the incentive to invest in upskilling programs [97,99]. Put differently, the higher research output in Europe and parts of Asia indicates stronger economic support for technological research and professional development initiatives.

4.5.5. *Weak policy and regulatory support*

Government policies and national strategies play a pivotal role in driving technological advancement [95,100]. The absence of clear policies or incentives to encourage AI adoption in construction can further exacerbate the skills gap. Regions with higher contributions, such as Europe, are likely to benefit from government-backed digital transformation agendas and supportive regulatory frameworks. Conversely, regions with no representation in this study, like North America and Africa, may lack such policy drivers, resulting in slower progress toward developing relevant AI competencies.

4.5.6. *Industry fragmentation and conservatism*

The construction industry is traditionally characterized by fragmentation and conservatism, which can slow the adoption of innovative practices [98,104]. In many developing regions, this conservatism is more pronounced due to a higher reliance on informal practices and small-scale contractors who may not see immediate value in investing in AI skills [111].

5. Discussion of review findings

This study's findings underscore the growing academic interest in AI-related skills within the construction industry, albeit with a fragmented and uneven global distribution. The upward trend in publications, particularly from Europe and Asia, reflects increasing awareness of the importance of digital competencies. However, the absence of contributions from Africa and North America reveals a significant research and practice gap that could hinder global progress in AI integration. There are more publications from Europe and Asia than Australia/Oceania and South America. Country of United Kingdom stands out in the AI-related skills in construction research. However, studies from continents such as Africa and North America that focus on this field are seemingly lacking, hence suggesting a significant research gap therein. This trend indicates a lack of academic or institutional focus on integrating AI competencies within the construction industry, which may hinder technological

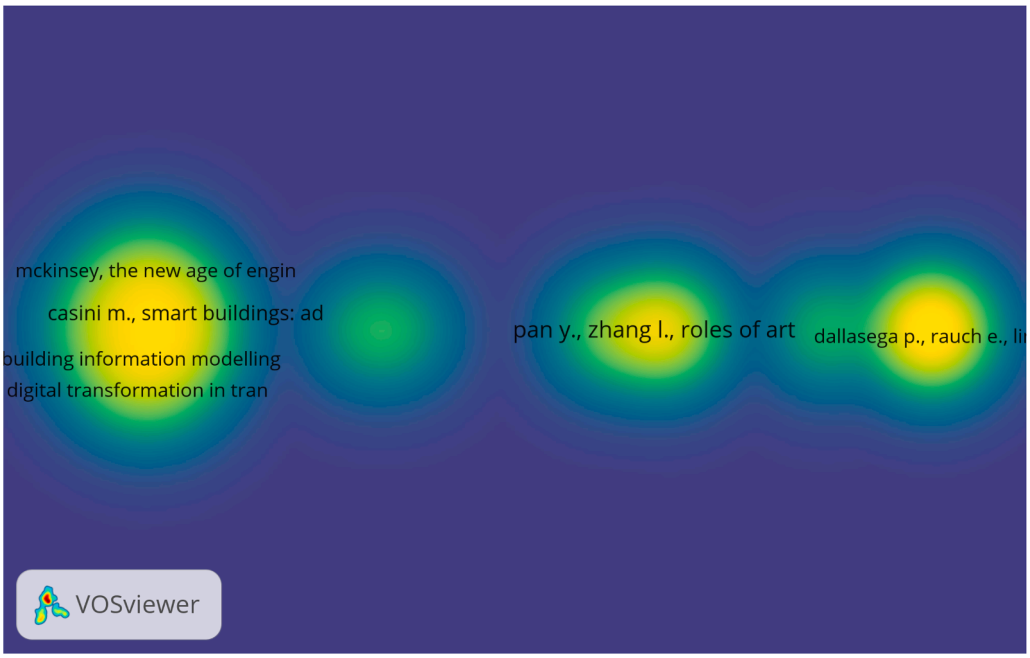

Fig. 8. Document co-citation network of AI-related skills in construction publications.

advancement and innovation. Also, it highlights the need for increased investment in research and capacity building to ensure global inclusivity in the digital transformation of the construction industry. Generally, the pace of research in AI-related skills within the construction sector is slow globally, reflecting limited academic engagement and practical implementation. This finding is consistent with the study by Souza and Debs [47], which highlights that only a limited number of investigations have focused on the skills necessary for engaging with emerging technologies, including AI, within the construction sector. This sluggish progress may be attributed to challenges like resistance to technology-driven change, insufficient funding, and lack of expertise within the construction industry (Parekh and Mitchell, 2024). Moreover, the findings from this study reveal that, the underdevelopment of AI skills among construction professionals is shaped by a complex interplay of educational, cultural, infrastructural, economic, policy-related, and industry-specific factors. Studying AI-related skills in construction is essential for equipping the industry with the technological expertise needed to improve innovation, efficiency, and productivity in an increasingly digital world (Adebowale and Agumba, 2023).

Drawing on the UTAUT, this study hypothesises that the development of AI skills among construction professionals is influenced by four critical constructs: performance expectancy, effort expectancy, social influence, and facilitating conditions. Specifically, low performance expectancy (i.e., perceiving AI as having limited benefit for job performance) and high effort expectancy (i.e., perceiving AI as difficult to learn) may discourage skill development. Similarly, limited social influence (i.e., lack of encouragement or role models within the organisation) and insufficient facilitating conditions (i.e., lack of resources, training programs, and infrastructure) can further hinder AI upskilling. By articulating each construct in detail, this study extends the utility of UTAUT from general digital skills adoption to the specific context of AI upskilling in construction. By integrating these four constructs, this study provides a comprehensive explanation of the persistent gaps in AI skill development among construction professionals. This articulation not only clarifies the barriers but also demonstrates the broader applicability and utility of the UTAUT framework in the context of AI-specific skill adoption. Consequently, it highlights the study's novelty and its contribution to both theory and practice. It contributes to current knowledge by positioning UTAUT as a valuable theoretical lens for understanding and addressing the unique challenges associated with AI upskilling in the construction industry. In practical terms, this refined understanding guides the development of targeted interventions, such as customised training programs, organisational policies to incentivize AI learning, and the establishment of robust support structures. Ultimately, such strategies can collectively enhance the readiness and capability of construction professionals to engage with AI technologies effectively.

The visualised author network confirms that AI-related skills in construction research across the globe remains in its early stages. It reveals only two established clusters, with most researchers dispersed across the network: indicating a significant lack of collaboration. In addition, the network shows that majority of recent contributions originate from these isolated researchers rather than from cohesive research groups. The co-occurring keywords primarily revolve around themes of construction professionals and the application of AI, mirroring the focus of the cited references. The co-occurrence of keywords and citation patterns further indicates a focus on AI applications rather than the human competencies required for their effective use.

Research in this domain tends to adopt theoretical frameworks, as evidenced by the use of terms such as 'information theory' and frequent citations of works by Selin and Rossi [112] and Pracucci [60]. This contrasts with the findings of Collins *et al.* [113], who highlighted the absence of theoretical frameworks in AI-related research. Authors such as David John Edwards and Qian Zhang are among the most frequently cited globally. The geographical diversity of these highly cited authors demonstrates that AI-related skills in construction research have a widespread global presence [114].

While the literature predominantly focuses on theoretical frameworks, promoting the practical application of AI tools and effectively translating these theoretical frameworks into actionable practice within the construction industry, requires the implementation of several strategic measures can be implemented. First, the development of industry-focused training programs and hands-on workshops is essential to equip professionals with practical AI skills beyond conceptual understanding. Additionally, the introduction of pilot projects and case studies that demonstrate the successful integration of AI frameworks into real-world construction processes can provide valuable evidence and learning opportunities. Establishing simulation and sandbox environments where practitioners can safely experiment with AI tools further encourages skill

Table 7
Framework of AI-related skills for construction professionals.

No	Skill Domain	Specific AI-Related Skills	Frequency	AEC Role Players
1	Data Literacy and Management	Data collection, preparation, and visualisation BIM data integration with AI tools	9	Project Managers, Engineers, BIM Specialists
2	AI and ML Fundamentals	Understanding AI/ML concepts Interpreting AI model outputs Application of predictive analytics	11	Site Engineers, Project Managers, Quantity Surveyors
3	Programming and Computation	Basic Python/R programming Logical thinking and automation setup	6	AI/BIM Coordinators, Tech-savvy Engineers
4	Digital Collaboration Tools	Use of cloud-based platforms (e.g., Procore, Revizto) AI-embedded project management software	7	Project Managers, Coordinators
5	AI-Driven Decision Making	Using AI insights for scheduling, cost control, and risk mitigation	3	Quantity Surveyors, Planners, Project Directors
6	Cybersecurity Awareness	Understanding AI-related risks Data privacy and protection protocols	3	IT Officers, Project Managers
7	Change Management and Adaptability	Managing digital transformation Adapting to AI adoption cycles	5	Senior Managers, HR/Training Units
8	Communication and Collaboration	Translating AI outputs into practical instructions Working with AI/IT experts	2	All professionals

Note: The eight skill domains presented in this table were derived through thematic grouping of the sixteen AI-related skills listed in Table 2 using content analysis.

development without project-related risks. Moreover, creating clear guidelines and toolkits derived from theoretical models can help bridge the gap between academic insights and practical implementation. Strengthening collaborations between academia and industry through joint research initiatives, knowledge exchange platforms, and co-developed training curricula can foster a more seamless adoption of AI innovations. Finally, supportive policies and incentive schemes from government and professional bodies can motivate construction firms to invest in AI integration and continuous professional development, ultimately enhancing the sector's overall technological capability and workforce competence.

The results from the meta-analysis of the 17 selected studies reveal the dominance of AI and ML fundamentals skill domain, hence reflecting the significance of this skill domain for leveraging AI to enhance productivity and innovation during construction project delivery. This is followed by data literacy and management skill domain. The AI and ML fundamentals skill domain include an understanding of AI/ML concepts, interpreting AI model outputs, and application of predictive analytics. Hence, there is a need for an understanding of AI and ML concepts which equip construction professionals with the foundational knowledge needed to engage with intelligent systems used in construction project delivery. This involves recognising how AI models function, what data they require, and the types of outputs they generate. Additionally, knowledge of interpreting AI model outputs is crucial for making informed decisions based on forecasts, risk assessments, or resource optimisation insights provided by AI tools [89]. Furthermore, application of predictive analytics allows construction professionals to anticipate project challenges like cost overruns, safety risks, or delays, thereby enhancing planning accuracy and proactive decision-making (Assaad et al., 2020). Fig. 9 presents the summary of the AI skills development in the construction industry.

Identified competencies such as data literacy, AI fundamentals, programming and computation align closely with those outlined in the Turing framework and CIOB framework [33,39]. Competencies like change management and collaboration resonate with the CIOB emphasis on incremental adoption, however, less frequently addressed. However, other critical domains such as AI-driven decision making, cybersecurity awareness, change management and adaptability, and communication and collaboration were less frequently addressed, suggesting underexplored areas in current research necessitating further exploration.

6. Study's implications and limitations

6.1. Implications of the study's findings

The findings of this study carry significant implications for the construction industry, particularly in the context of digital transformation and workforce development. As AI technologies become increasingly embedded in construction processes, there is a pressing need for professionals to acquire new competencies that go beyond traditional technical skills. The identification of eight core AI-related skill domains, ranging from data literacy and AI fundamentals to cybersecurity and change management, highlights the multifaceted nature of digital readiness. These competencies are essential not only for operating AI tools but also for interpreting their outputs, managing digital workflows, and fostering collaborative, data-driven decision-making environments.

For industry practitioners, the findings of this study suggest the rethinking of professional development strategies to incorporate both hard and soft AI skillsets. As such, construction firms must invest in upskilling their workforce to ensure that employees can effectively engage with AI systems and contribute to smarter, more efficient project delivery. Similarly, educational institutions and training providers need to revise curricula to reflect the evolving demands of the industry. Interdisciplinary programs that blend construction management and related built environment disciplines with data science and digital technologies remain crucial in preparing the next generation of professionals.

From a policy perspective, the study underscores the need for frameworks that support AI competency development, particularly in underrepresented regions such as Africa and North America, where research and training initiatives remain limited. Policymakers and professional bodies should promote inclusive strategies that ensure equitable access to AI education and resources, thereby enabling global participation in the digital transformation of construction.

In addition, the research community is encouraged to address the evident gaps in current literature, particularly in areas such as ethical AI use, cybersecurity, and change management. More collaborative and practice-oriented studies are needed to build a cohesive understanding of how AI skills can be effectively developed and applied in real-world construction settings. By bridging these gaps, the industry can better harness the transformative potential of AI and ensure that its workforce is equipped to thrive in an increasingly digital and data-driven environment.

Furthermore, this study advances theoretical understanding by extending the UTAUT to the context of AI skills in construction. While UTAUT emphasises performance expectancy, effort expectancy, social influence, and facilitating conditions as key determinants of technology adoption, the study's findings show how these constructs directly

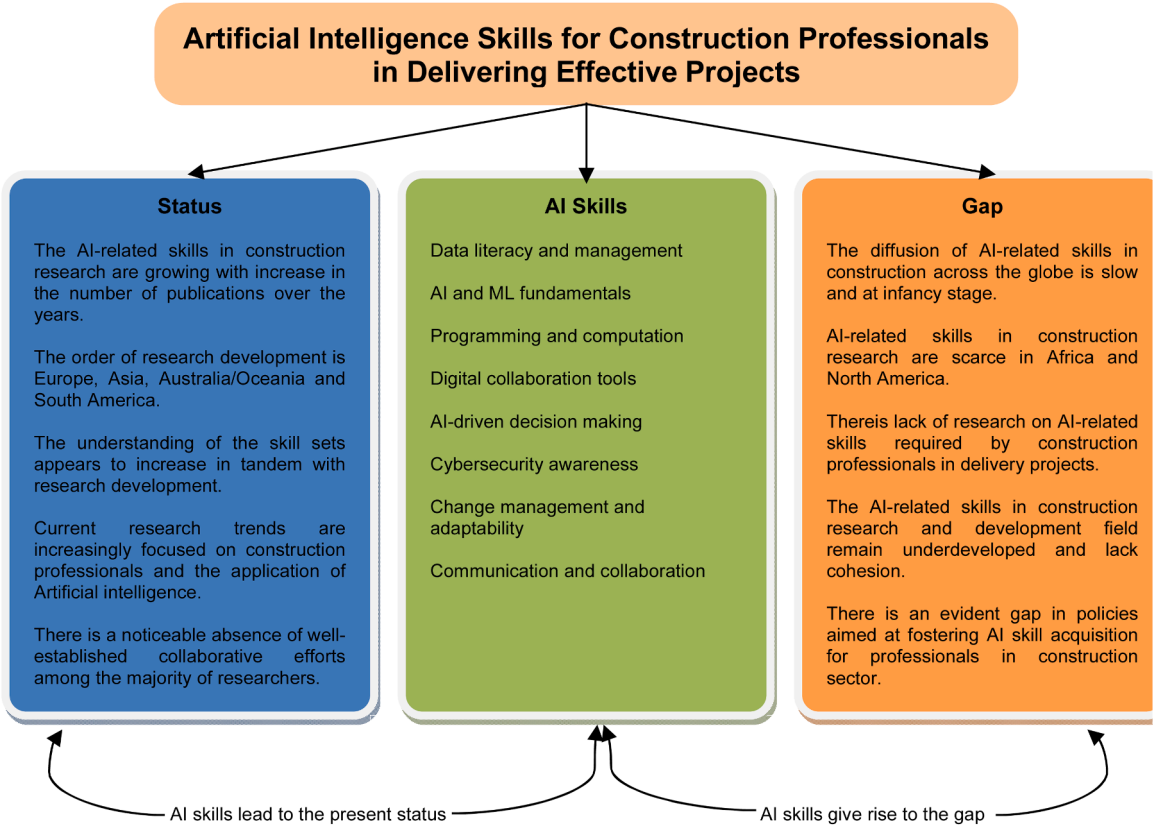

Fig. 9. AI skills development for construction professionals in delivering effective projects.

translate into specific AI-related skill domains required by construction professionals. By linking performance expectancy to skills that enhance efficiency and decision-making, effort expectancy to abilities that simplify AI interactions, social influence to organisational and collaborative competencies, and facilitating conditions to training and infrastructure support, this study enriches UTAUT by providing a detailed, skill-based perspective. This theoretical integration offers a holistic framework for understanding and developing AI competencies, thereby fostering more effective and sustainable AI adoption in construction project delivery.

6.2. Study's limitations and directions for future research

While this review has primarily focused on the people aspect, particularly skills and competencies required for AI adoption in the construction industry, it is important to acknowledge that successful implementation of AI depends on a diverse set of interconnected factors. Beyond skills, there are other critical elements that interact to influence AI integration and uptake. First, technological tools and infrastructure represent an essential component. The availability, accessibility, and maturity of AI-related tools, such as IoT devices, machine learning algorithms, advanced data analytics platforms, and robotics, significantly affect an organisation's ability to adopt AI effectively [113,115]. Without adequate technological readiness, even a highly skilled workforce may encounter challenges in deploying AI solutions. Secondly, processes and workflows play a very important role. Well-defined, streamlined, and adaptable processes enable the smooth integration of AI into existing project delivery methods [116]. Fragmented or rigid workflows can hinder the adoption process, resulting in operational inefficiencies and reduced value realisation. Therefore, construction organisations need to assess and redesign processes to be compatible with AI-driven approaches. And thirdly, organisational governance and leadership commitment are fundamental enablers. This includes strategic alignment of AI initiatives with organizational goals, establishment of clear policies and standards, provision of adequate resources, and fostering an innovation-supportive culture [117]. Effective governance ensures that AI adoption is not only a technological or operational decision but a strategically guided transformation effort [118].

Therefore, in light of the above information regarding other critical factors that interact to influence AI adoption, *future research directions* should adopt a more holistic perspective by examining the interplay among skills, technological tools, processes, and governance structures. Such comprehensive investigations can provide a more nuanced understanding of barriers and enablers to AI adoption and inform tailored strategies for different contexts within the construction industry. By integrating these additional dimensions, future studies can contribute to developing a robust framework that supports the practical and sustainable implementation of AI in construction, ultimately enhancing productivity, safety, and project outcomes.

7. Conclusions

The growing call for an AI-enhanced built environment sector is reshaping how construction projects are planned, managed, and delivered. However, implementing AI in construction does not rely solely on technical tools; it also depends on the human skills that enable professionals to understand, apply, and critically evaluate these tools in practice. As the construction sector faces increasing complexity and a growing demand for data-driven decision-making, the need to clearly understand, define, and cultivate AI-related skills among construction professionals has become urgent.

Meanwhile, there are factors leading to the underdevelopment of AI skills in the construction industry. This study reveals that the underdevelopment of AI skills among construction professionals is shaped by a complex interplay of educational, cultural, infrastructural, economic,

policy-related, and industry-specific factors. Addressing these underlying factors is essential for equipping construction professionals globally with the necessary skills to leverage AI technologies effectively, thereby enhancing productivity, innovation, and competitiveness across the sector.

This study conducted a scientometric review and meta-analysis of AI-related skills in construction research; an area that remains underrepresented in the current literature. By tracing the global evolution of research on AI competencies, it identified core themes, influential contributors, and knowledge gaps in how construction professionals are being prepared for digital transformation. The analysis revealed a growing body of work addressing AI in construction, yet a fragmented and limited focus on AI-specific skills.

From the analysis, eight core skills were synthesized: Data Literacy and Management, AI and ML Fundamentals, Programming and Computation, Digital Collaboration Tools, AI-Driven Decision Making, Cybersecurity Awareness, Change Management and Adaptability, and Communication and Collaboration. These skills were mapped against two contemporary AI skills frameworks to identify alignment and gaps between academic research and professional practice. The comparison revealed substantial overlap in areas such as digital literacy and AI/ML fundamentals, while also highlighting underexplored areas such as ethical stewardship, AI-driven decision-making and reflective evaluation.

This review offers practical insights into the skills needed for AI readiness. By bridging scientometric evidence with practical competency frameworks, the study provides both conceptual clarity and applied relevance. It underscores the urgent need to equip construction professionals with the skills required to thrive in an AI-driven built environment. The research presents a timely roadmap for skills development relevant to both professionals and construction organizations.

Finally, the study acknowledges the growing need to integrate traditional construction expertise with both soft and hard AI skills. Organizations must consider how to incorporate AI skill models into professional development strategies to ensure that site workers and project professionals are knowledgeable and equipped to work with, and alongside intelligent systems. Particular attention should be given to areas where research may be falling short, such as AI-driven decision-making, cybersecurity awareness, change management and adaptability, and communication and collaboration. This study is limited by its reliance on published literature indexed in selected databases, which may exclude relevant industry reports, grey literature, or non-English publications. Additionally, while the scientometric and meta-analytical approach provides a broad overview, it may not capture the nuanced, context-specific skill requirements across different regions or construction sub-sectors. Future research could benefit from qualitative investigations and stakeholder interviews to validate and expand upon the identified skill sets.

CRediT authorship contribution statement

Lovelin Ifeoma Obi: Writing – original draft, Data curation. **Innocent Chigozie Osuizugbo:** Writing – original draft, Methodology, Data curation. **Bankole Osita Awuzie:** Writing – review & editing, Writing – original draft, Conceptualization.

Declaration of competing interest

There is no conflict of interest.

Data availability

Data will be made available on request.

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